JOURNAL CLUB / 生物精神病学与发育神经科学

孤独症谱系障碍社会行为的神经环路

从机制到干预 | From Mechanism to Intervention

Neural Circuits of Social Behavior in Autism Spectrum Disorder

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行为发育科文献汇报 · 主讲约 25 分钟 · 基于用户上传原文

文献定位：跨物种证据整合的叙述性综述

项目	原文信息与解读
作者与机构	Columbia / NYSPI； Cincinnati Children’s
版本与日期	Journal Pre-proof； 2026-10-02 接收
证据来源	人类影像、啮齿类环路、干预、发育研究
研究类型	叙述性整合；不提供合并效应量

评价重点：构念是否清晰、跨尺度推断是否成立、临床转化仍缺什么。

研究问题：广义社会症状可能遮蔽不同机制

临床与试验层面

01 干预效应有限、个体差异显著

02 任务改善难以泛化至日常生活

03 广义诊断／社会总分作为主要终点

作者提出的机制问题

01 相同社会表现是否由不同限制产生？

02 不同过程依赖哪些部分可分离环路？

03 发育时点如何改变领域之间的耦合？

引文： Sandbank 2023 BMJ [4]； Carruthers 2020 Autism Res [6]； Sikich 2021 NEJM [7]。

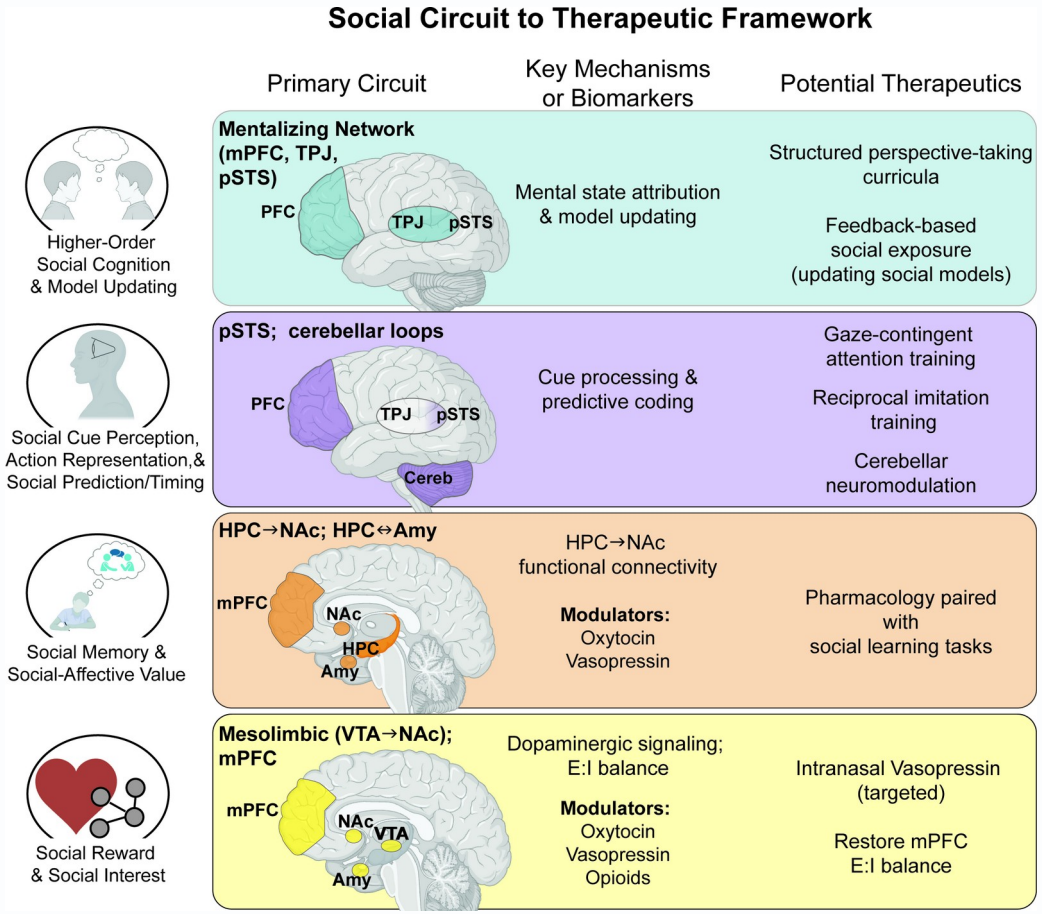
Table 1： 四个功能—计算领域及其构念边界

原文领域	核心计算	构念边界
1 社会奖赏与兴趣	价值赋予／启动参与	接近行为 ≠ 想要 ≠ 喜欢
2 社会记忆与情感价值	识别熟人／赋予情感意义	注意、编码、记忆需分辨
3 线索／动作／预测与时序	动态取样、映射、预测	行动理解 ≠ 心理状态归因
4 高阶认知与模型更新	信念推断／修订人物模型	非单一 ToM； 受任务要求影响

与 RDoC 社会过程构念有重叠，但本框架用于细分机制、发育轨迹和领域交互。

PDF p.5–6； Table 1 p.31 （整理）

Figure 1： 从功能领域连接候选环路与干预



阅读图示的三个前提

1. 顺序非层级（not hierarchical）

2. 领域部分可分离，同时相互作用

3. 候选治疗 ≠ 已验证临床有效

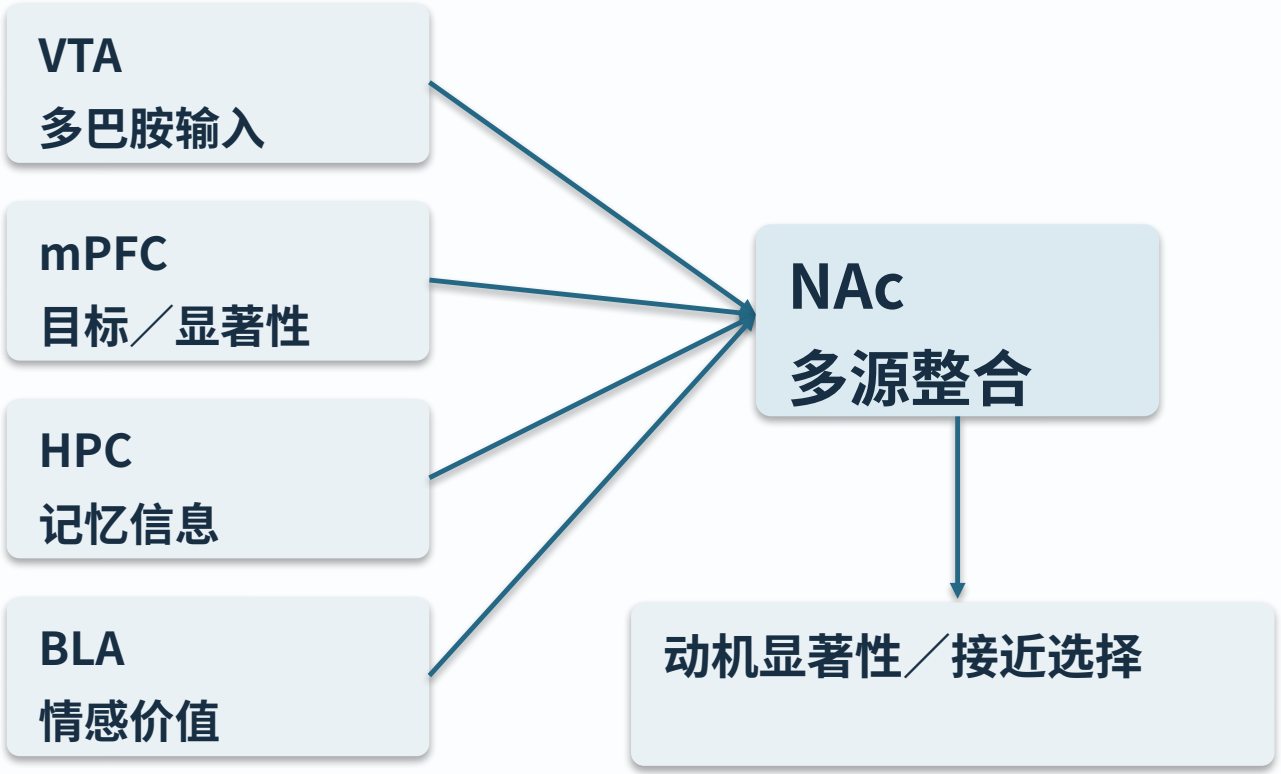
保留原图框架；后续环路示意均为据正文整理，箭头不一律代表单突触投射。

I 社会奖赏与兴趣： wanting 与 liking 必须区分

构念	计算／调节	可观察行为的解释限制
Wanting	激励显著性；多巴胺与学习	少发起可由多种限制产生
Liking	享乐体验；阿片等情感系统	不等同于互动次数
社会兴趣	寻求／维持互动的动机	ASD 内部异质性显著
竞争奖赏	局限兴趣可能有较强价值	非全局奖赏系统低反应

Berridge & Kringelbach 2015 Neuron [23]； Clements 2018 JAMA Psychiatry [25]。

VTA–NAc 环路：将多源输入整合为动机显著性



据原文整理；仅绘示本文讨论的关系，非解剖学定位图。

人类关联

VTA–NAc 结构连接与社会困难相关 [26]

动物机制

VTA→NAc 多巴胺投射参与社会奖赏 [28]

刺激类别

局限兴趣与社会奖赏可呈不同反应 [38]

Supekar 2018 Brain [26]； Gunaydin 2014 Cell [28]； Iyer 2025 Nat Commun [29]。

mPFC 与社会兴趣：细胞类型及 E:I 证据的转化边界

层级	原文证据	解释范围
啮齿类环路	丘脑– mPFC； E:I； D2R+ 投射细胞	模型内社会偏好与兴奋性
人体研究	遗传／尸检／ MRS ／ fMRI 建模	E:I 推断间接、方法依赖
转化假设	抑制张力／同步性作为候选指标	用于分层或靶点参与研究
尚缺证据	可靠预测治疗反应并改善功能	不能由 E:I 异常直接选药

Brumback 2018 Mol Psychiatry [43]； Trakoshis 2020 eLife [45] 。

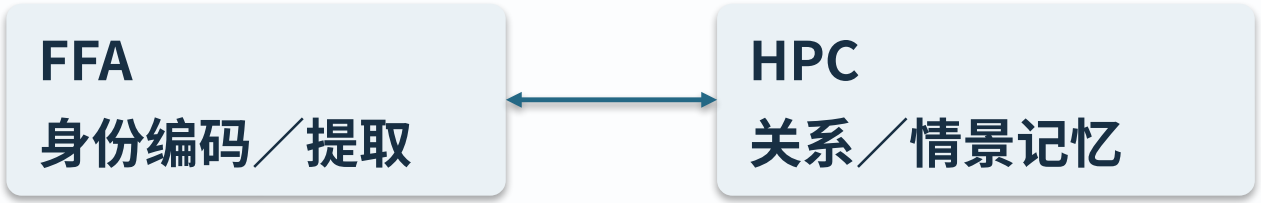
神经肽转化：机制依据与大型阴性试验并存

证据	代表研究（原文编号）	综述结论
机制支持	Dölen 2013 Nature [32]	NAc 催产素与 5-HT 协同参与社会奖赏
大型儿科 RCT	Sikich 2021 NEJM [7]	鼻内催产素：核心社会结局阴性
大型儿科 RCT	Hollander 2022 JAMA Psychiatry [8]	Balovaptan 拮抗剂：社会结局阴性
初步信号	Parker 2019 [35]； Guastella 2023 [36]	小试验／幼龄信号，需独立验证

机制可解释候选方向；不能替代目标人群、给药方案与临床主要终点的验证。

II 社会记忆：身份识别与关系／情景信息协同

人类功能分工（协同，不表示直接投射）



啮齿类环路证据



据原文整理；仅绘示本文讨论的关系，非解剖学定位图。

人类证据

身份／情景记忆差异及 HPC–PCC 关联 [51]

动物证据

CA2； dCA2→vCA1； HPC→NAc [57–61]

构念控制

区分注意取样、知觉编码、巩固与提取

Hitti 2014 Nature [57]； Meira 2018 Nat Commun [59]； Okuyama 2016 Science [61]。

社会情感价值： HPC–BLA 与 ACC–BLA 的不同作用



据原文整理；仅绘示本文讨论的关系，非解剖学定位图。

HPC ↔ BLA

既往互动的情感联结与社会接近 [65]

ACC → BLA

社会来源厌恶信息；观察性恐惧学习 [66]

证据边界

动物投射操控 ≠ 已确立的人体病因或疗效

Felix-Ortiz & Tye 2014 J Neurosci [65]； Allsop 2018 Cell [66]。

记忆领域的转化逻辑：可塑性增强需配对学习

机制与候选策略

01 海马神经肽调节：加压素／催产素

02 配对有意义、重复的社会经验

03 增强编码／提取并促进泛化

临床证据与限制

01 D- 环丝氨酸＋社会技能训练小试验

02 疗效尚未确立；需足够规模复制

03 同时评价近端过程和家校功能

Minshawi 2016 [62]； Wink 2017 [63]； Aye 2021 Cochrane [64]。

PDF p.10–11； Table 1； refs.58、 60、 62–64

III 实时社会加工：三个部分可分离的子过程

子过程	候选系统	主要计算
社会线索感知	pSTS；与杏仁核／ PFC 交互	动态凝视、表情、生物运动
动作表征	感觉运动／动作观察网络	外部动作→内部动作表征
预测与时序	小脑– PFC ／颞顶闭环	预测、误差校正、互动同步

动态输入处理为高阶推断提供基础，但不等同于信念、意图等心理状态归因。

PDF p.11–13； Table 1 （整理）

pSTS： 动态社会线索异常与可调节性的证据

研究类型	原文代表引文	支持的结论
人体任务研究	Pelphrey 2005 Brain [72]	ASD 凝视变化加工存在差异
Meta 分析	Todorova 2019 Mol Autism [73]	跨范式的生物运动知觉差异
幼儿干预	Wang 2020 Autism Res [74]	眼动依赖提示可改变注意取样
训练／可行性研究	Faja 2012 [75]； Pereira 2019 [77]	系统可调节；长期泛化待检验

应区分：任务参与改变、脑活动调节、领域获益，以及日常功能获益。

动作表征：运动困难可降低共同活动中的表征精度

01

观察

外部动作／运动线索

02

表征

映射至自身动作库

03

协调

识别目的与互动同步

04

学习

互惠模仿／共同活动

ASD： 协调与感觉反应差异可削弱实时表征。

互惠模仿训练：共同注意发起及社会情绪功能改善 [85] 。

Ingersoll 2012 J Autism Dev Disord [85]；动作理解与高阶心理状态归因需区分。

预测与时序：小脑机制及跨疾病外推的限度

机制依据

01 预测误差与先验权重配置

02 小脑–前额叶／颞顶闭环

03 时序、误差校正与人际同步

证据边界

01 小脑结构差异不等于特异生物标志物

02 刺激研究来自先天小脑畸形人群

03 ASD 有效性、安全性与分层需另行检验

Pellicano & Burr 2012 [86]； Kelly 2021 [87]； Oldrati 2024 Neuroimage Clin [89]。

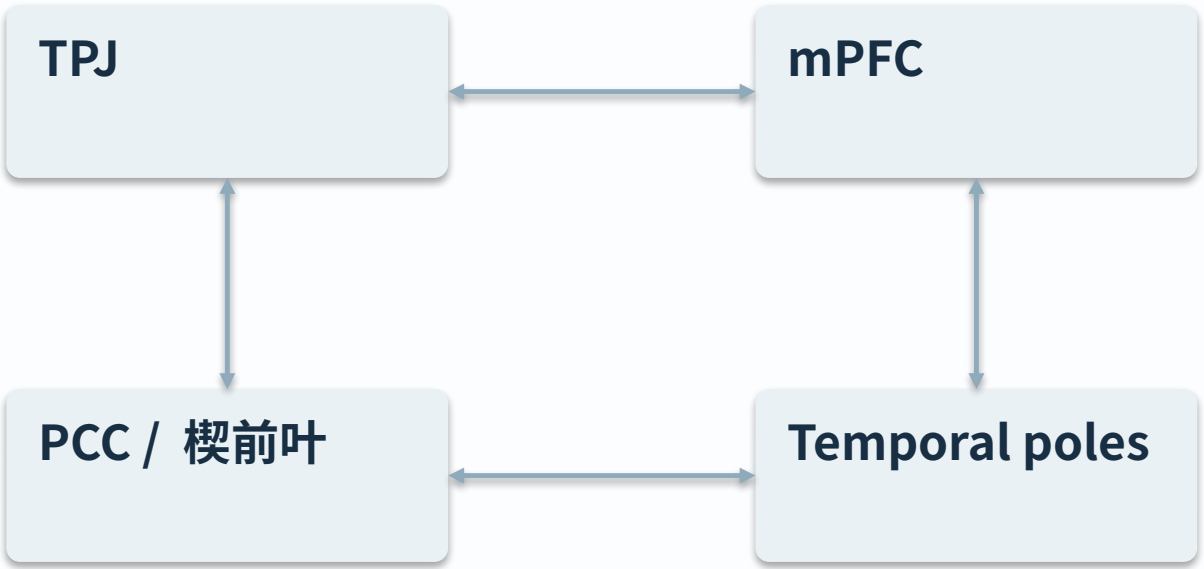
IV 高阶社会认知：显性推断、自发推断与模型更新

子过程	心理计算	关键依赖
显性心理状态归因	有意识推理信念、欲望、意图	语言、执行、情境整合
自发社会推断	时间压力下实时归因	动态线索取样与网络协同
社会模型更新	依据经验修订他人模型	心理化网络与强化学习系统

同一领域内部也可出现分离；上游输入或学习差异可改变高阶任务表现。

PDF p.14–16； Table 1 （整理）

显性归因：心理化网络与任务要求共同决定表现



pSTS： 动态线索输入

据原文整理；仅绘示本文讨论的关系，非解剖学定位图。

网络

TPJ 、 mPFC 、 PCC ／ precuneus 、 temporal poles

任务要求

情境整合、语言与执行影响所测差异 [97]

转化

结构化观点采择需配对自然情境练习

Saxe & Kanwisher 2003 [95]； Padmanabhan 2017 [96]； Gao 2023 [97] 。

自发推断与模型更新：从“会推理”到“实时使用”

自发社会推断

01 pSTS–TPJ–mPFC 动态交互

02 显性任务通过 ≠ 自发追踪完整

03 线索可获取性与时间压力是关键

社会模型更新

01 心理化网络＋强化学习价值系统

02 ACC 信息流在 vmPFC 汇合 [100]

03 显性反馈逐渐撤除：待检验假设

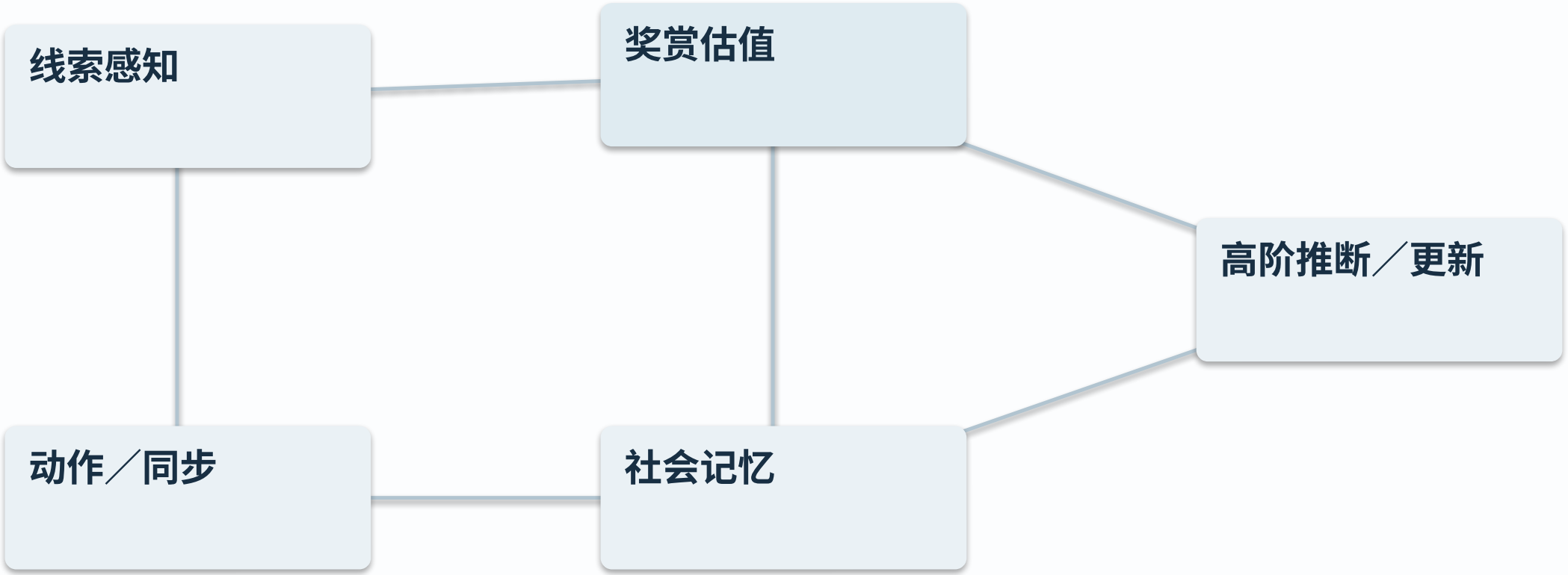
Senju 2012 Neuroscientist [98]； Behrens 2008 Nature [100] （典型成人证据）。

发育证据：系统成熟时程并不同步

	婴儿期	儿童期	青春期
社会注意／奖赏	出生面孔偏好； 2–6 月眼注视轨迹 [50]		
记忆／身份加工	FFA： 5–7 岁可见活动，特异性延续至青春期 [67]		
动态加工／时序	6 月 pSTS 互动选择性 [90]；小脑持续儿童期成熟		
心理化／更新	7 月 TPJ 功能组织 [101]； 5–11 岁特异性增强 [104]		

上述为不同研究的发育观察；不能据此直接确定个体治疗窗口。

发育级联：任一领域都可能成为早期限制



纵向干预检验：某一领域改变，是否先于并预测跨领域变化？

Cruz-Sanchez 2026 Cell Rep [110]：路径敏感期动物证据；具体 ASD 社会因果级联仍属初步。

Box 1： 社会接近、共情与攻击行为跨越四领域

跨领域现象	原文核心限定	临床／研究后果
社会接近质量	数量不能直接读出兴趣／奖赏	分解时机、互惠、距离与情境
共情／亲社会行为	非统一缺失；受组分与测量影响	分测情感／认知／行动
攻击行为	非单一 ASD 特异环路	先医学与功能评估，再匹配支持

Box 1 扩展框架边界：相似外显行为可能由不同生物与环境状态汇聚而来。

Box 2： 环路知情的领域特异治疗开发示例

01	**临床表型**	定向／发起少；局限兴趣竞争强化
02	**领域假设**	线索显著性与奖赏引导学习受限
03	**候选测量**	眼动；期待 / 结果任务； VTA–NAc ； pSTS
04	**学习干预**	自然、回应性、有意义的社会互动
05	**生物学辅助**	富集亚组＋主动学习；研究性检验
06	**结局检验**	近端改变 → 家校参与；纵向测级联

原文注释： research-development example ， not a current clinical algorithm 。

学术评价：框架贡献明确，验证路径仍有缺口

本文的框架贡献

01 细分社会过程及其构念边界

02 整合环路、发育与学习机会

03 提出富集样本与机制一致终点

尚需验证的关键问题

01 叙述性选材：系统性与偏倚不确定

02 跨物种／跨尺度与因果级联未闭环

03 ASD 特异性、代表性与预测效度不足

近期用途：假设生成、试验富集和靶点参与；尚不足以按脑区直接匹配治疗。

PDF p.3–4 、 p.18–20 ； 汇报者依据原文评价

结论：领域表型 × 发育时点 × 机制一致的结局

WHAT 明确受限过程，避免社会总分掩盖异质性

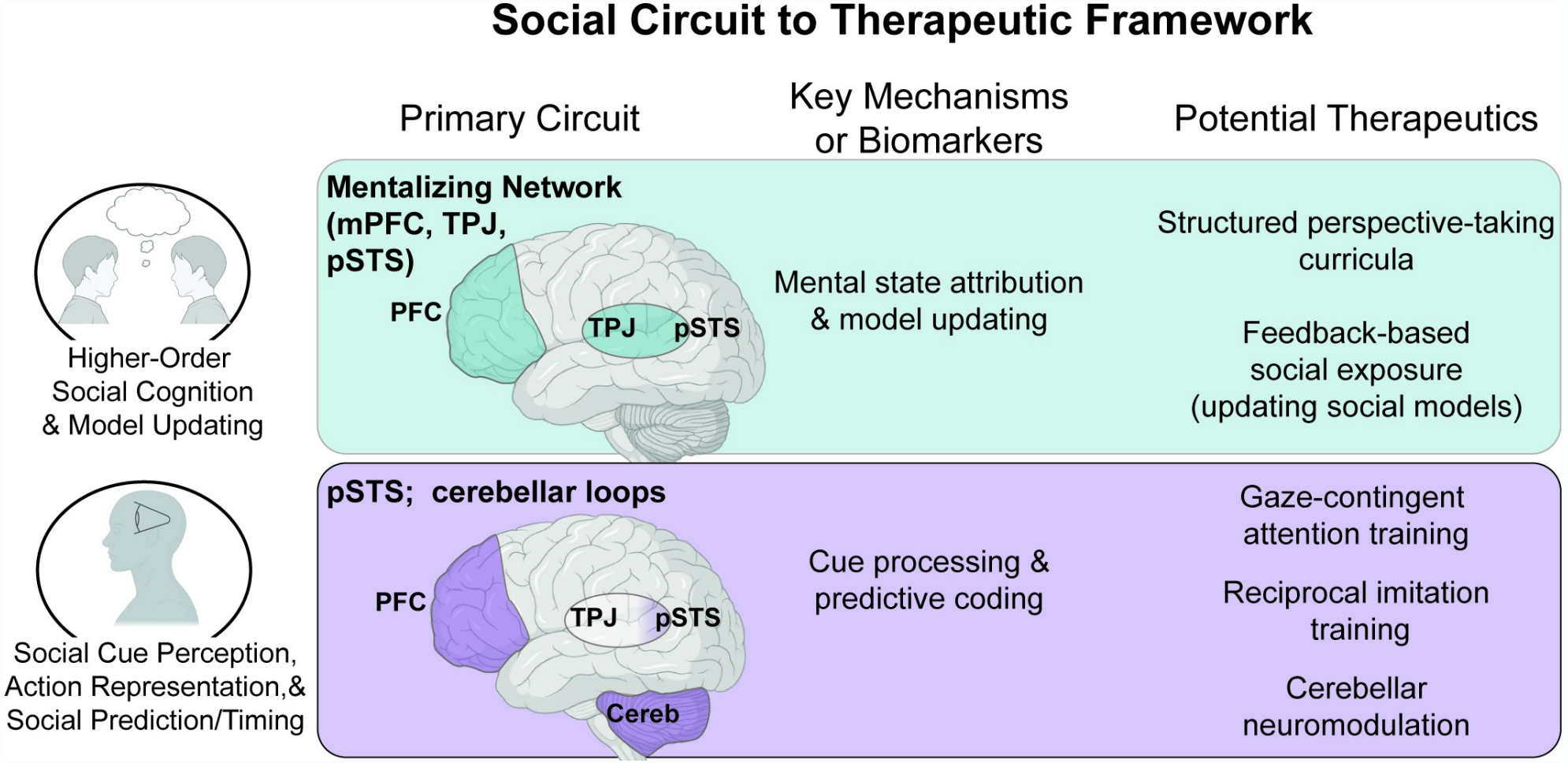
WHEN 检验发育时点与跨领域级联

HOW 富集亚组、靶点参与、主动学习与真实泛化

讨论：哪一个领域指标最值得先验证其“超出常规评估”的增量价值?

PDF p.17–19；原文结论整理

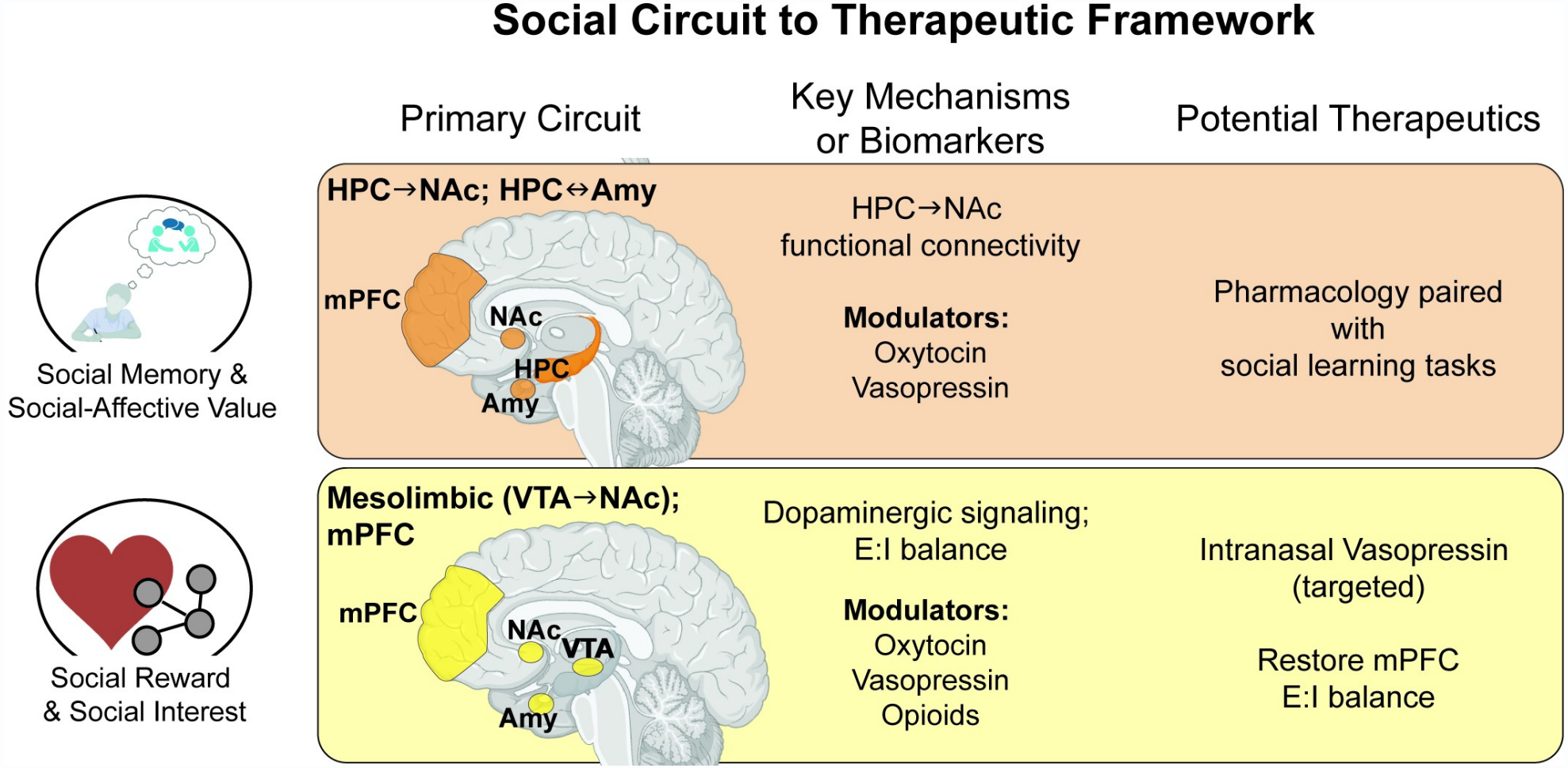
备查 | Figure 1 上半部：高阶认知与动态加工



原图局部；保留原始列标题。其余两领域见下一页。

PDF p.32 Fig.1 （上半部局部放大）

备查 | Figure 1 下半部：社会记忆与奖赏



原图下半部；上方补入同图原始列标题，未改动科学内容。

PDF p.32 Fig.1 （下半部局部放大）

备查｜Table 1：定义、边界与 RDoC 对应

Table 1. Behavioral domain definitions and boundaries

1. Social Reward and Social Interest	
Definition	The appetitive value of social contact and the motivation to initiate or maintain social interaction.
Boundary	Distinguishes wanting/incentive salience from liking/hedonic impact. Social approach may reflect reward and interest, but observed approach is not a direct measure of either construct.
Related RDoC Constructs	Affiliation and Attachment Social Communication
Translational Implication	Identify whether reduced, atypical, or competing social reward constrains learning; pair reward modulation with active social learning opportunities.
2. Social Memory and Social-Affective Value	
Definition	Recognition of familiar individuals and learned emotional valence attached to people, voices, faces, and contexts.
Boundary	Face memory can reflect attention, perceptual encoding, or memory-specific processes; these should be assessed separately.
Related RDoC Construct(s)	Perception and Understanding of Others Affiliation and Attachment Social Communication
Translational Implication	Strengthen encoding and retrieval by aligning cue salience, repetition, and emotionally meaningful learning contexts.
3. Social Cue Perception, Action Representation, and Social Prediction/Timing	
Definition	Dynamic processing of gaze, facial expression, biological motion, timing, and action kinematics; prediction of what others will do next; mapping observed actions onto motor repertoires.
Boundary	Provides input to higher-order inference but is not equivalent to mental-state attribution.
Related RDoC Construct(s)	Social communication Perception and Understanding of Others.
Translational Implication	Assess and target cue sampling, imitation, motor synchrony, and prediction errors while testing whether gains generalize to naturalistic interaction and higher-order social cognition.
4. Higher-Order Social Cognition and Model Updating	
Definition	Explicit and spontaneous inferences about beliefs, intentions, emotions, traits, reliability, and social norms; updating person models over time.
Boundary	Not a unitary ToM deficit and not fully cross-species; task performance depends on language, executive function, context integration, and cue availability.
Related RDoC Construct(s)	Social Communication Perception and Understanding of Self Perception and Understanding of Others
Translational Implication	Test explicit inference and model-updating strategies while ensuring adequate perceptual input and opportunities for naturalistic practice and generalization.

原表完整保留；会议主讲使用第 4 页的可编辑中文整理表。

PDF p.31 Table 1 （完整表）

备查 | Box 1： 社会接近、共情与攻击

Box 1. Special Considerations: Quality of Social Approach, Empathy, and Aggression

Quality of Social Approach: The quality of social approach in ASD involves the manner, timing, and reciprocity, and contextual fit of how a person interacts with others, more than simply the amount of contact. Profiles can include reduced initiation, context-specific avoidance, active-but-odd or intrusive approach, interpersonal-distance differences, poor timing/synchrony (25, 115, 116). The observed approach should not be treated as a direct readout of social interest or reward because it may reflect different combinations of cue salience, valuation, anxiety or threat, sensory load, motor planning, communication demands, and prior experience (25, 115, 117). No single “abnormal approach network” has been established. Candidate systems include mesolimbic and ventromedial/orbitofrontal prefrontal circuits for incentive value and initiation (117); amygdala–insula–cingulate and regulatory prefrontal systems for threat appraisal and avoidance; pSTS–fusiform–amygdala networks for gaze and dynamic social-cue processing (118); dorsal intraparietal–fusiform–amygdala coupling for interpersonal-space regulation (119); and cerebellar and sensorimotor systems for timing and interpersonal synchrony (120). Frontostriatal and orbitofrontal control systems are plausible contributors to intrusive or poorly inhibited approach, but direct ASD-specific evidence is less mature.

Empathy: Empathy is not a unitary process or a single-circuit function. Affective empathy or the sharing of, or resonance with, another person’s emotional state, reliably engages anterior insula and anterior/mid-cingulate systems, whereas cognitive empathy and mental-state inference recruit the temporoparietal junction, mPFC, posterior superior temporal sulcus, precuneus/posterior cingulate cortex, and temporal poles (121-123). These systems interact with valuation and response-selection processes when empathic understanding is translated into helping, generosity, or other prosocial behavior. In autism, the evidence supports heterogeneity rather than a uniform absence of empathy: findings vary with the component assessed, task format, social context, measurement method, and co-occurring alexithymia (124, 125). Studies of spontaneous helping and generosity likewise show that prosocial behavior cannot be inferred from a single global empathy score (14–15). Empathy and prosocial behavior are therefore best treated as cross-domain outcomes that depend on social-cue perception, affective salience and resonance, mental-state inference, valuation, learning, and selection of a response.

Aggression: Behavior directed toward another person or property that could cause harm is clinically important in ASD, but it is neither a diagnostic criterion nor a unitary social-processing domain. The function and precipitants of aggression should be assessed before assigning a mechanism. Contributors may include pain or medical illness, sleep disruption, anxiety, sensory overload, disrupted routines, communication barriers, and reinforcement contingencies (126-128). The general neuroscience literature describes distributed defensive and aggression-related circuitry involving the amygdala, medial hypothalamus, and periaqueductal gray, with prefrontal, anterior cingulate, and striatal systems modulating context, frustration, and inhibitory control (126). Direct evidence for an ASD-specific aggression circuit is limited; the most defensible formulation is that several biological and environmental states can converge on the same observable behavior. However, it is possible that activation of aggression circuitry coincides with activation of frustration circuitry, which includes the anterior cingulate cortex, ventral striatum, VTA and the PFC. Thus, overloading or under-activation, of the PFC for behavioral inhibition may result in lower frustration tolerance and increased aggression.

Clinical implications for aggression: Management should begin with medical assessment and functional behavioral assessment, followed by function-based behavioral and communication intervention when indicated (118, 128). Risperidone and aripiprazole have randomized-trial evidence and regulatory approval for irritability associated with autistic disorder, which encompasses aggression, tantrums, and self-injury (129). Whenever possible the treatment target should be the identified driver of the behavior, such as pain, anxiety, sleep disturbance, communication frustration, or reinforcement contingency, rather than nonspecific suppression of an assumed aggression circuit. In addition, ameliorating the underlying frustration, such as by providing alternative communication methods, may also offer alternative approaches to reducing aggression.

原框完整保留；中文要点与临床边界见第 22 页。

PDF p.33 Box 1 （完整原文）

备查 | Box 2：研究开发示例，不是临床算法

Box 2: Worked Example of Circuit-informed, Domain-Specific Treatment Development

Clinical profile	Child shows low spontaneous orientation to faces/voices, limited initiation of social bids, and strong competing reinforcement from restricted interests. This profile is not interpreted as low social enjoyment until anxiety, sensory sensitivity, motor planning, language, and prior social adversity are assessed.
Domain hypothesis	Primary constraint may involve social cue salience and reward-guided learning, with secondary effects on social memory and later mentalizing because fewer contingent social experiences are sampled.
Candidate biomarker measures	Eye tracking during naturalistic scenes; caregiver-reported social motivation/affiliation and unusual approach scales; reward tasks separating anticipation from outcome; structural/functional VTA–NAc connectivity and task-related pSTS engagement.
Intervention/therapeutic logic	Begin with naturalistic, developmentally appropriate social engagement that increases contingent, predictable, and intrinsically meaningful social input. Use restricted interests strategically to bridge into social interaction. Add gaze-contingent cueing, imitation, or parent-mediated approaches when cue sampling is a bottleneck.
Adjunctive biological strategy	A pharmacological or neuromodulatory adjunct should be tested only in an enriched subgroup and paired with active learning. For example, neuromodulation of social salience or learning would be hypothesized to show greatest benefit when paired with active social-learning opportunities while the child is engaged in social learning opportunities.
Outcome test	Measure proximal change in cue sampling, reward learning, and social initiation, followed by generalization to home/school participation. Test cascade hypotheses longitudinally rather than inferring them from cross-sectional associations.

Note: This is a research-development example, not a current clinical algorithm

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BLA / ACC / pSTS	基底外侧杏仁核 / 前扣带 / 后上颞沟
TPJ / mPFC / E:I	颞顶联合区 / 内侧前额叶 / 兴奋 : 抑制
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